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- 3D point clouds are often noisy and cluttered. Many downstream applications are sensitive to the quality of input points.

- We present BaySurf-SANF, a novel end-to-end pipeline for explicit surface reconstruction from noisy point clouds.

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- We reformulate the task of generating surface points into a simpler problem: transforming randomly sampled points to the shape surface.

- Since clutter points greatly impact any geometric processing, we break down our procedure into two steps: **Remove Clutter** and **Sample Surface**.

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- [illegible]

We utilize an affine dynamic layer (ADL) to enhance the flexibility of transformations of NFs. The ADL is computed as follows:

$$\begin{aligned} y_a &= \exp(s_1(C)) \odot x_a + t_1(C) \\ y_b &= \exp(s_2(y_a, C)) \odot x_b + t_2(y_a, C) \\ y &= \text{concat}(y_a, y_b), \end{aligned}$$

where x and y : the input and output; C : the conditional feature; D : the dimension of x or y ($D = a + b$); s and t : functions for scale and translation.

- Training objective of the flow-based network in the second step.

$$\mathcal{L}_{\text{total}} = \alpha \mathcal{L}_{\text{prior}} + \beta \mathcal{L}_{\text{rec}} + \gamma \mathcal{L}_{\text{ne}},$$

- $\mathcal{L}_{\text{prior}}$: the negative log-likelihood to optimize the transformation capability of NFs.
- $\mathcal{L}_{\text{rec}}$: a similarity measure between the generated points and the ground truth, computed using the Chamfer distance.
- $\mathcal{L}_{\text{ne}}$: a penalty of the deviation of the estimated normals from the ground truth normals.

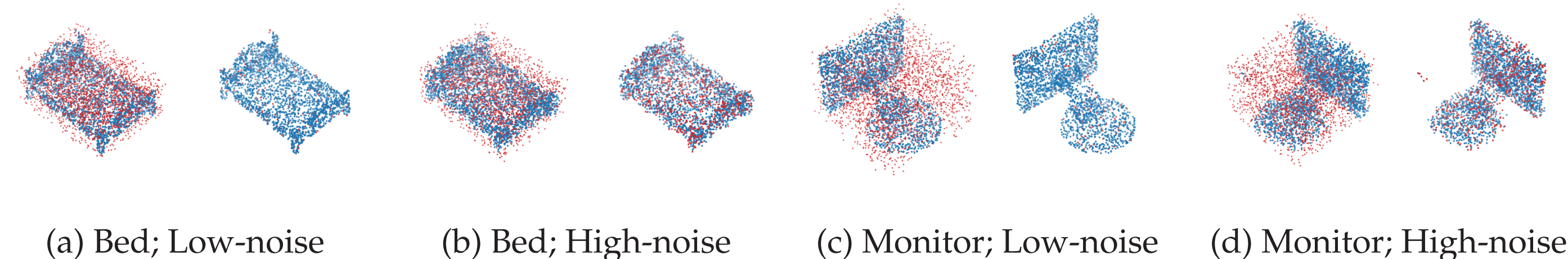
- Finally, we combine the output pair (**generated points** and **estimated normals**) using a post-processing step, Screened Poisson Surface Reconstruction (SPSR), to produce a high-quality 3D mesh of the underlying surface.

- **ModelNet10 dataset.** To generate noisy and cluttered point clouds:

1. **Sampling:** Given a total point count n , we use the Trimesh library to sample n_s surface points from each mesh.
2. **Noise injection:** We add Gaussian noise to these surface points at two levels of standard deviation.
3. **Clutter addition:** The remaining $n_c = n - n_s$ points are sampled uniformly within the bounding box to simulate clutter.
4. **Normalization:** Each noisy, cluttered point cloud is centered at the origin and scaled to fit within a unit sphere.

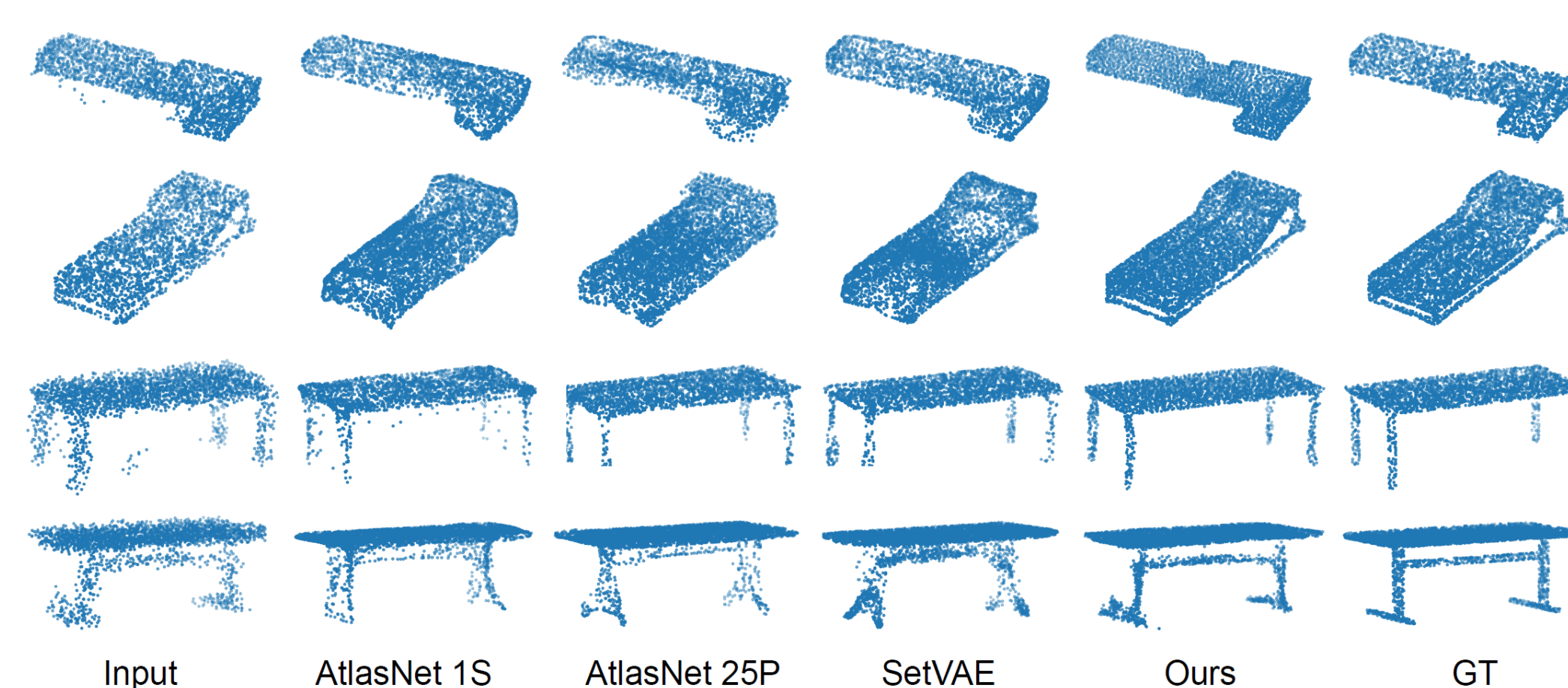
- Results of Step 1: Point Labeling

	Accuracy	Precision	Recall
Low-noise	0.9322	0.8997	0.9728
High-noise	0.7730	0.7194	0.8953



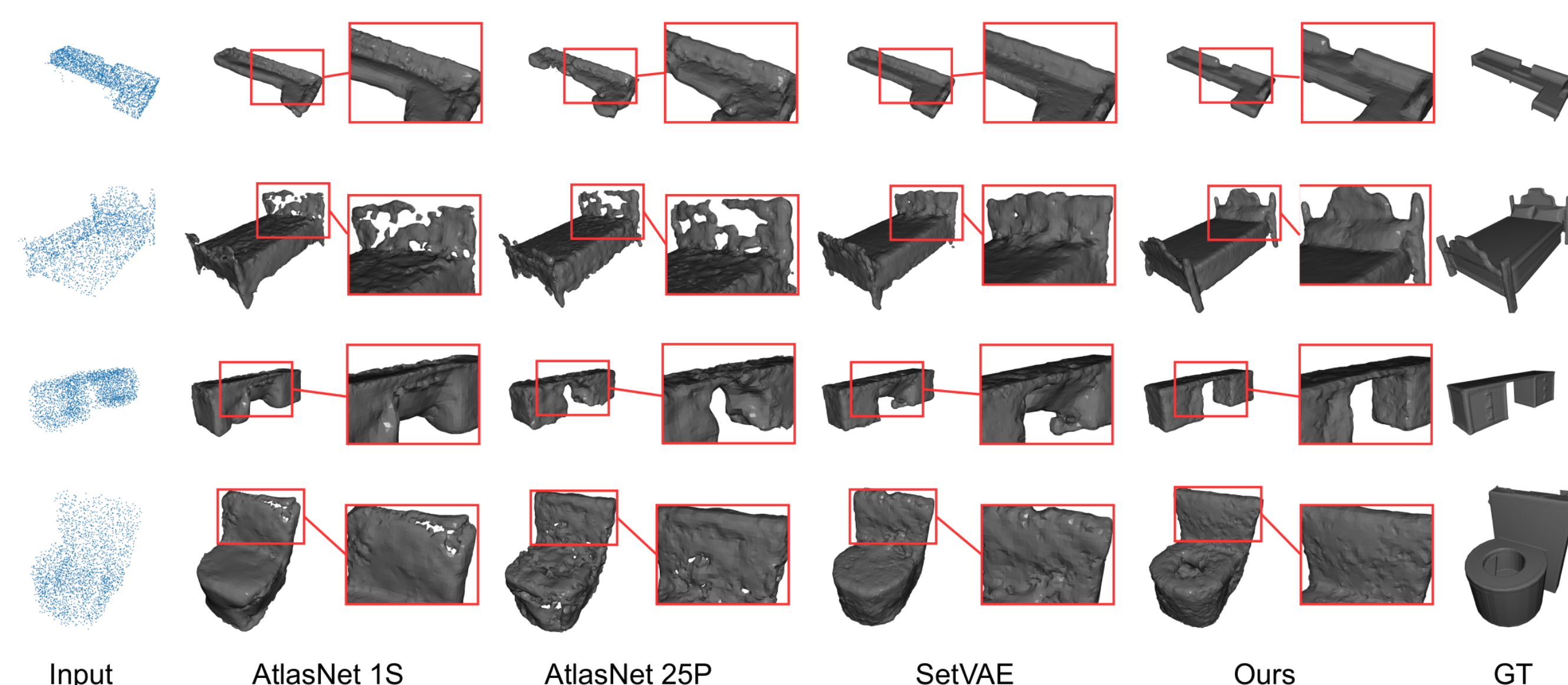
- Results of Step 2: Point Generation

	Low-noise		High-noise	
Method	2.5k pts	5k pts	2.5k pts	5k pts
AtlasNet 1S	3.09	2.29	3.04	2.39
AtlasNet 25P	2.54	1.79	2.61	1.88
SetVAE	2.17	1.43	2.40	1.66
Ours	1.20	0.74	1.69	1.21



- Results of Surface Reconstruction

	Low-noise, 2.5k pts				Low-noise, 5k pts			
	CD		EMD		CD		EMD	
Method	Mean	Med.	Mean	Med.	Mean	Med.	Mean	Med.
AtlasNet 1S	0.0347	0.0008	0.0341	0.0007	0.0512	0.0008	0.0505	0.0008
AtlasNet 25P	0.2682	0.0006	0.2794	0.0006	0.2868	0.0005	0.2937	0.0006
SetVAE	0.0134	0.0005	0.0131	0.0005	0.0091	0.0004	0.0091	0.0005
Ours	0.0553	0.0002	0.0580	0.0003	0.0063	0.0002	0.0075	0.0003
	High-noise, 2.5k pts				High-noise, 5k pts			
	CD		EMD		CD		EMD	
Method	Mean	Med.	Mean	Med.	Mean	Med.	Mean	Med.
AtlasNet 1S	0.0405	0.0008	0.0396	0.0008	0.1609	0.0009	0.1664	0.0010
AtlasNet 25P	0.2699	0.0006	0.2949	0.0007	0.0147	0.0006	0.0140	0.0007
SetVAE	0.0154	0.0006	0.0145	0.0006	0.0142	0.0005	0.0137	0.0006
Ours	0.0081	0.0003	0.0073	0.0004	0.0069	0.0003	0.0068	0.0004



- Ablation Study (Model 7: Complete Model)

									Low-noise, 2.5k pts				Low-noise, 5k pts			
S.H. M.H.				NFs NFs NFs				CD		EMD		CD		EMD		
Model	Att.	Att.	LVs	NE	ACL	AIL	ADL	Mean	Med.	Mean	Med.	Mean	Med.	Mean	Med.	
1	-	✓	✓	-	-	-	-	0.2945	0.0012	0.3030	0.0014	0.0233	0.0008	0.0221	0.0010	
2	✓	-	-	-	-	-	✓	0.0575	0.0002	0.0598	0.0003	0.0060	0.0002	0.0072	0.0003	
3	-	✓	-	-	-	-	✓	0.0567	0.0002	0.0592	0.0003	0.0514	0.0002	0.0531	0.0003	
4	-	✓	✓	-	-	-	✓	0.0560	0.0002	0.0565	0.0003	0.0045	0.0002	0.0055	0.0003	
5	-	✓	✓	✓	✓	-	-	0.0571	0.0002	0.0588	0.0003	0.0062	0.0002	0.0073	0.0003	
6	-	✓	✓	✓	-	✓	-	0.0554	0.0002	0.0610	0.0003	0.0598	0.0002	0.0639	0.0003	
7	-	✓	✓	✓	-	-	✓	0.0553	0.0002	0.0580	0.0003	0.0063	0.0002	0.0075	0.0003	

									High-noise, 2.5k pts				High-noise, 5k pts			
S.H. M.H.				NFs NFs NFs				CD		EMD		CD		EMD		
Model	Att.	Att.	LVs	NE	ACL	AIL	ADL	Mean	Med.	Mean	Med.	Mean	Med.	Mean	Med.	
1	-	✓	✓	-	-	-	-	0.2120	0.0014	0.2153	0.0018	0.0456	0.0009	0.0458	0.0011	
2	✓	-	-	-	-	-	✓	0.0392	0.0005	0.0389	0.0006	0.1245	0.0006	0.1280	0.0007	
3	-	✓	-	-	-	-	✓	0.0186	0.0004	0.0182	0.0005	0.0113	0.0004	0.0111	0.0005	
4	-	✓	✓	-	-	-	✓	0.0110	0.0004	0.0104	0.0004	0.0080	0.0003	0.0079	0.0004	
5	-	✓	✓	✓	✓	-	-	0.1623	0.0004	0.1668	0.0004	0.0116	0.0003	0.0120	0.0004	
6	-	✓	✓	✓	-	✓	-	0.0293	0.0004	0.0316	0.0004	0.0083	0.0003	0.0078	0.0004	
7	-	✓	✓	✓	-	-	✓	0.0081	0.0003	0.0073	0.0004	0.0069	0.0003	0.0068	0.0004	